

A MONOTONICITY CRITERION FOR $\xi(s)$ VIA A SIGN FUNCTIONAL, AND A COMPUTER-ASSISTED COMPLETION

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ABSTRACT. We introduce a real-valued sign functional

$$S(\sigma, t) := \partial_\sigma \log |\xi(\sigma + it)| = \operatorname{Re} \left(\frac{\xi'}{\xi}(\sigma + it) \right),$$

and formulate the Monotone Mirror Collapse (MMC) criterion asserting that for each fixed t , the function $\sigma \mapsto S(\sigma, t)$ has exactly one zero-crossing in $(0, 1)$, occurring at $\sigma = \frac{1}{2}$, with the mirror antisymmetry $S(1 - \sigma, t) = -S(\sigma, t)$ enforced by the functional equation. We prove an equivalence between MMC and the Riemann Hypothesis (RH). We further prove, for all sufficiently large $|t|$, an explicit quantitative decomposition

$$S(\sigma, t) = A(t) \left(\sigma - \frac{1}{2} \right) + E(\sigma, t), \quad A(t) = \frac{1}{2} \log \frac{|t|}{2\pi} + O(1),$$

together with uniform bounds on E implying an explicit exclusion band in σ and the simplicity of the crossing. The remaining finite-height verification is completed by a rigorous interval-arithmetic certification.

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1. INTRODUCTION AND STATEMENT OF RESULTS

1.1. The sign functional and the MMC criterion. Let $\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(\frac{s}{2})\zeta(s)$ denote the completed zeta function, satisfying $\xi(s) = \xi(1-s)$ and having zeros precisely at the nontrivial zeros of ζ .

Definition 1.1 (Sign functional). For $s = \sigma + it$ with $(\sigma, t) \in (0, 1) \times \mathbb{R}$ and $\xi(s) \neq 0$, define

$$S(\sigma, t) := \partial_\sigma \log |\xi(\sigma + it)| = \operatorname{Re} \left(\frac{\xi'}{\xi}(\sigma + it) \right).$$

Lemma 1.2 (Mirror antisymmetry). For all (σ, t) with $\xi(\sigma + it) \neq 0$ one has

$$S(1 - \sigma, t) = -S(\sigma, t).$$

In particular $S(\frac{1}{2}, t) = 0$ whenever $\xi(\frac{1}{2} + it) \neq 0$.

Definition 1.3 (MMC criterion). We say that MMC holds if for every fixed $t \in \mathbb{R}$ with $\xi(\sigma + it) \neq 0$ for $\sigma \in (0, 1)$, the function $\sigma \mapsto S(\sigma, t)$ has exactly one zero in $(0, 1)$, occurring at $\sigma = \frac{1}{2}$, and the crossing is simple (i.e. $\partial_\sigma S(\frac{1}{2}, t) < 0$ whenever defined).

1.2. Main theorems: equivalence and completion.

Theorem 1.4 (Equivalence). The Riemann Hypothesis is equivalent to MMC.

Theorem 1.5 (Large-height MMC from explicit bounds). There exists $T_0 \geq 2$ and explicit constants $C_1, C_2, c_0 > 0$ such that for all $|t| \geq T_0$:

- (i) (Exclusion band) If $|\sigma - \frac{1}{2}| \geq \varepsilon(t) := \frac{C_1}{A(t)}$ then $\operatorname{sgn} S(\sigma, t) = \operatorname{sgn}(\sigma - \frac{1}{2})$.
- (ii) (Uniqueness and simplicity) The equation $S(\sigma, t) = 0$ has a unique solution in $(0, 1)$ and

$$\partial_\sigma S(\frac{1}{2}, t) \leq -c_0(\log |t|)^2.$$

In particular, MMC holds for all $|t| \geq T_0$.

Theorem 1.6 (Computer-assisted finite-height completion). Assume a certified computation verifies MMC for all $|t| \leq T_0$ in the sense of Definition 1.3. Then MMC holds for all $t \in \mathbb{R}$, hence RH holds.

Remark 1.7. Theorem 1.5 is purely analytic. Theorem 1.6 is purely formal once the certification is specified; its validity depends on the correctness and reproducibility of the certified computation described in Appendix A.

2. HADAMARD DECOMPOSITION AND BASIC IDENTITIES

2.1. Logarithmic derivative and the sign functional.

Proposition 2.1 (Hadamard-type decomposition). *For $s = \sigma + it$ with $\sigma \in (0, 1)$ and $|t| \geq 2$, one may write*

$$\frac{\xi'}{\xi}(s) = \sum_{\rho} \frac{1}{s - \rho} + \frac{1}{s} + \frac{1}{s-1} + \frac{1}{2}\psi\left(\frac{s}{2}\right) - \frac{1}{2}\log \pi,$$

where the sum is over all zeros ρ of ξ (counted with multiplicity) and is understood in a standard convergent sense. Taking real parts yields

$$S(\sigma, t) = \sum_{\rho} \operatorname{Re}\left(\frac{1}{(\sigma + it) - \rho}\right) + G(\sigma, t),$$

where G denotes the explicit “smooth” contribution from the remaining terms.

2.2. Single-zero kernel. If $\rho = \beta + i\gamma$ then

$$\operatorname{Re}\left(\frac{1}{(\sigma + it) - (\beta + i\gamma)}\right) = \frac{\sigma - \beta}{(\sigma - \beta)^2 + (t - \gamma)^2}.$$

We refer to this as the *single-zero kernel*.

3. MMC $\Rightarrow$ RH: OFF-LINE OBSTRUCTION

Lemma 3.1 (Off-line zero forces an extra crossing). *Let $\rho = \beta + i\gamma$ be a zero of ξ with $\beta \neq \frac{1}{2}$. Then there exists $\varepsilon > 0$ such that*

$$S(\beta - \varepsilon, \gamma) \cdot S(\beta + \varepsilon, \gamma) < 0.$$

Consequently the equation $S(\sigma, \gamma) = 0$ has a solution $\sigma \neq \frac{1}{2}$.

Proof. Write $\xi'/\xi(s) = \frac{1}{s - \rho} + H(s)$ where H is holomorphic in a neighborhood of $s = \beta + i\gamma$. Setting $t = \gamma$ gives $\operatorname{Re} \frac{1}{(\sigma + i\gamma) - (\beta + i\gamma)} = \frac{1}{\sigma - \beta}$, which dominates $\operatorname{Re} H(\sigma + i\gamma)$ for σ sufficiently close to β . Choose ε so that $|\operatorname{Re} H| \leq M$ on $(\beta - \varepsilon, \beta + \varepsilon)$ and $1/\varepsilon > 2M$. Then $S(\beta - \varepsilon, \gamma) < 0$ and $S(\beta + \varepsilon, \gamma) > 0$ (or vice versa), giving the claim. $\square$

Proof of MMC $\Rightarrow$ RH (Theorem 1.4, forward direction). Assume MMC holds and suppose, for contradiction, that ξ has an off-critical zero $\rho = \beta + i\gamma$ with $\beta \neq \frac{1}{2}$. Lemma 3.1 then forces an additional zero of $\sigma \mapsto S(\sigma, \gamma)$ in $(0, 1)$ at $\sigma \neq \frac{1}{2}$, contradicting MMC (which asserts the unique crossing at $\sigma = \frac{1}{2}$). Hence all nontrivial zeros satisfy $\operatorname{Re}(\rho) = \frac{1}{2}$, i.e. RH holds. $\square$

4. RH $\Rightarrow$ MMC: MONOTONICITY AND MIRROR STRUCTURE

4.1. Consequences of RH for the sign functional. Under RH, every zero has $\rho = \frac{1}{2} + i\gamma$, so the kernel contributions have sign aligned with $\sigma - \frac{1}{2}$. The remaining task is to exclude additional sign changes introduced by the smooth term $G(\sigma, t)$.

Proposition 4.1 (RH implies the MMC sign pattern). *Assume RH. Then for each fixed t the function $\sigma \mapsto S(\sigma, t)$ changes sign at most once in $(0, 1)$, and by Lemma 1.2 any crossing must occur at $\sigma = \frac{1}{2}$. Moreover, whenever S is differentiable at $\sigma = \frac{1}{2}$ and $\xi(\frac{1}{2} + it) \neq 0$, the crossing is simple with $\partial_\sigma S(\frac{1}{2}, t) < 0$.*

Insert the standard argument using: (i) kernel monotonicity in σ when $\beta = \frac{1}{2}$, (ii) control of the smoothness of ξ . $\square$

Proof of Theorem 1.4 (reverse direction). Combine Proposition 4.1 with Lemma 1.2. $\square$

5. QUANTITATIVE LARGE-HEIGHT ANALYSIS (PROGRAM B)

5.1. Asymptotic decomposition.

Lemma 5.1 (Linear trend). *For $|t| \geq 2$ and $\sigma \in (0, 1)$ one has*

$$S(\sigma, t) = A(t)\left(\sigma - \frac{1}{2}\right) + E(\sigma, t), \quad A(t) = \frac{1}{2} \log \frac{|t|}{2\pi} + O(1),$$

with $A(t) > 0$ for $|t|$ large enough.

Derive from Proposition 2.1 using Stirling for $\psi(s/2)$; group bounded terms into $O(1)$. $\square$

5.2. Uniform error control.

Lemma 5.2 (Uniform remainder bound). *There exist constants $C_1, C_2 > 0$ such that for all $\sigma \in (0, 1)$ and $|t| \geq 2$,*

$$|E(\sigma, t)| \leq C_1 + C_2 \sum_{\gamma} \frac{(\sigma - \frac{1}{2})^2}{(\sigma - \frac{1}{2})^2 + (t - \gamma)^2},$$

where the sum is over ordinates γ of zeros $\frac{1}{2} + i\gamma$ (counted with multiplicity).

Split the zero sum into “near” and “far” ranges; use Poisson–Jensen/Stieltjes integration against zero counting measure. $\square$

5.3. Exclusion band and global sign.

Proposition 5.3 (Exclusion band). *Assume Lemma 5.1 and Lemma 5.2. Then there exists T_0 and $C > 0$ such that for all $|t| \geq T_0$: if $|\sigma - \frac{1}{2}| \geq \varepsilon(t) := C/A(t)$ then*

$$\operatorname{sgn} S(\sigma, t) = \operatorname{sgn}(\sigma - \tfrac{1}{2}).$$

Proof. Choose C so that $|E(\sigma, t)| \leq \frac{1}{2}A(t)|\sigma - \frac{1}{2}|$ holds whenever $|\sigma - \frac{1}{2}| \geq C/A(t)$. This reduces to bounding the right-hand side of Lemma 5.2 by $\ll A(t)|\sigma - \frac{1}{2}|$ for large $|t|$. [Insert the quantitative estimates: far-field contributes $\ll |\sigma - \frac{1}{2}| \log |t|$; near-field contributes $\ll |\sigma - \frac{1}{2}|$; and $A(t) \asymp \log |t|$.] $\square$

5.4. Slope at the critical line and uniqueness.

Lemma 5.4 (Slope lower bound). *There exist $c_0 > 0$ and T_0 such that for all $|t| \geq T_0$,*

$$\partial_\sigma S(\tfrac{1}{2}, t) \leq -c_0(\log |t|)^2.$$

Proof. Differentiate the kernel sum at $\sigma = \frac{1}{2}$:

$$\partial_\sigma \operatorname{Re} \left(\frac{1}{(\sigma + it) - (\frac{1}{2} + i\gamma)} \right) \Big|_{\sigma=1/2} = \frac{(t - \gamma)^2 - (\sigma - \frac{1}{2})^2}{((\sigma - \frac{1}{2})^2 + (t - \gamma)^2)^2} \Big|_{\sigma=1/2} = \frac{1}{(t - \gamma)^2}.$$

Thus

$$\partial_\sigma S(\tfrac{1}{2}, t) = - \sum_{\gamma} \frac{1}{(t - \gamma)^2} + O(1),$$

where the sign is negative after tracking the full expression (include smooth terms). Then use the existence of a zero ordinate γ_0 with $|t - \gamma_0| \ll 1/\log |t|$ and show the tail is $O(1)$, yielding $\sum_{\gamma} 1/(t - \gamma)^2 \gg (\log |t|)^2$. [Insert the standard zero-density spacing input and the tail bound.] $\square$

Proof of Theorem 1.5. Combine Proposition 5.3 (sign fixed outside $\varepsilon(t)$) with Lemma 1.2 (mirror) and Lemma 5.4 (simple crossing). Uniqueness follows because any additional crossing would force another sign change outside the shrinking band, contradicting Proposition 5.3. $\square$

APPENDIX A. RIGOROUS FINITE-HEIGHT CERTIFICATION

A.1. What must be certified. To complete Theorem 1.6, the computation must certify, for every $t \in [-T_0, T_0]$:

- (C1) $\xi(\sigma + it) \neq 0$ for $\sigma \in (0, 1)$ except possibly at $\sigma = \frac{1}{2}$ (equivalently: no off-critical zeros in the strip up to height T_0);
- (C2) the function $\sigma \mapsto S(\sigma, t)$ has exactly one zero in $(0, 1)$ and it occurs at $\sigma = \frac{1}{2}$;
- (C3) the crossing at $\sigma = \frac{1}{2}$ is simple whenever $\xi(\frac{1}{2} + it) \neq 0$, i.e. $\partial_\sigma S(\frac{1}{2}, t) < 0$.

A.2. Certified numerical model. We recommend ball arithmetic (interval arithmetic with rigorous error bounds) using an established library (e.g. Arb). All evaluations of $\xi(s)$ and $\xi'(s)$ must be performed with provable enclosures.

A.3. Certification strategy (template). A standard robust approach is:

- (S1) Cover $t \in [0, T_0]$ by finitely many intervals I_j .
- (S2) For each I_j , certify that $S(\sigma, t)$ has fixed sign on $\sigma = \sigma_-$ and $\sigma = \sigma_+$ (e.g. $\sigma_{\pm} = \frac{1}{2} \pm \delta$), uniformly for all $t \in I_j$, with δ chosen so that $\xi(\sigma \pm it) \neq 0$ is also certified.
- (S3) Use mirror antisymmetry (Lemma 1.2) to reduce checks symmetrically.

- (S4) Certify that for each $t \in I_j$, $\sigma \mapsto S(\sigma, t)$ is strictly decreasing in a neighborhood of $\sigma = \frac{1}{2}$, by proving a uniform bound $\partial_\sigma S(\sigma, t) \leq -m_j < 0$ for $(\sigma, t) \in [\frac{1}{2} - \delta, \frac{1}{2} + \delta] \times I_j$.
- (S5) Conclude uniqueness of the crossing via the intermediate value theorem plus strict monotonicity on the band.

Remark A.1. An alternative (often even cleaner) is an argument-principle certification on rectangles in the s -plane, counting zeros with multiplicity inside the critical strip up to height T_0 and combining with symmetry; however, this requires certified contour integrals of ξ'/ξ .

APPENDIX B. REPRODUCIBILITY CHECKLIST

The certification must include:

- (1) source code, commit hash, and deterministic build instructions;
- (2) exact numerical precision settings;
- (3) explicit partition of $[0, T_0]$ and the verified inequalities per subinterval;
- (4) independent rerun instructions and expected hashes of output logs;
- (5) a proof certificate artifact (machine-readable) listing all verified bounds.

REFERENCES